

ICISSP 2026 - Paper #71

Paper Title: *Can Synthetic Spam Beat Real-World Detectors? Evaluating LLMs' Dual Role in Spam Generation and Detection*

Reviewer #1

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 5
Originality (Newness of the ideas expressed): 3
Technical Quality (Theoretical soundness/methodology): 4
Significance (Is the problem worth the given attention?): 5
Presentation (Structure/Length/English): 5
Overall Rating (Weighted value of above items): 4

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? No
Needs comparative evaluation? No
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? Yes
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

Notes to the Authors

This is a very interesting study, and I have wondered about the core theme myself. I hope the minor comments assist in the article revision process. Well done.
All the best.

Title of Reviewed Submission:

Can Synthetic Spam Beat Real-World Detectors? Evaluating LLMs' Dual Role in Spam Generation and Detection

Comments:

Title

– I think the title clearly reflects the objectives of the paper.

Abstract

– The abstract nicely summarises the paper and its findings. Well done.

Introduction

– The introduction is well written and poses some good questions to a relevant and persistent problem. I

agree with the contributions listed – especially number 5 which will enable future researchers to work on similar problems.

Background

- This is nicely written and summarises the background quite nicely without providing too much information.

Research Method

- These two sections are very detailed and the figures that follow are also impressive. For the most part it seems that the typical machine learning "best-practices" have been adhered to.
- My only question is around the evaluation metrics chosen. Why were these metrics chosen? A table summarising them and the motivation for each technique would enhance the paper in my opinion.

Results and Discussion

- The results are clear, and the results are interesting, especially which models performed better which is surprising. I would suggest more explicit future research recommendations to be made and that the limitations be separated out into its own section.

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Reviewer #2

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 6
Originality (Newness of the ideas expressed): 5
Technical Quality (Theoretical soundness/methodology): 5
Significance (Is the problem worth the given attention?): 5
Presentation (Structure/Length/English): 4
Overall Rating (Weighted value of above items): 5

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? No
Needs comparative evaluation? No
Improve critical discussion ? (validation): No
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? Yes
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The article examines the impact of Large Language Models (LLMs) in the context of spam generation and detection, adopting a dual perspective that considers both attackers and defenders. The study aims to assess how LLMs can be exploited to produce more sophisticated spam, as well as how they can be employed to develop more effective defense mechanisms.

Strengths

The topic is highly relevant and timely, addressing a pressing issue in cybersecurity and AI ethics. The authors provide a well-contextualized introduction that situates the study within the broader research landscape. Importantly, the datasets used in the experiments are made publicly available, ensuring transparency and reproducibility.

A notable strength lies in the dual evaluation of both offensive and defensive uses of LLMs. While much of the prior research has focused exclusively on defensive strategies, this paper acknowledges that attackers can adapt their methods once protective mechanisms are known. The methodology is clearly described and appropriate for the objectives of the study.

The results are of significant interest, and the Discussion section effectively highlights their implications and relevance. Moreover, the limitations of the work are explicitly stated, which adds to its credibility.

and academic rigor.

Weaknesses

Section 5 (Evaluation) is written in a somewhat telegraphic style, which hampers the smoothness and readability of the text. In addition, the distribution of tables between the main body and the Appendix makes it more difficult for readers to follow the experimental results cohesively.

Other comments

There are minor grammatical inaccuracies that should be corrected. For example, "As Figure 1 and Table 3 shown" should read "As Figure 1 and Table 3 show"; and in the sentence beginning "As Figure 3 and Table 4, training", a verb such as "show" appears to be missing.

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Reviewer #3

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 6

Originality (Newness of the ideas expressed): 5

Technical Quality (Theoretical soundness/methodology): 4

Significance (Is the problem worth the given attention?): 4

Presentation (Structure/Length/English): 4

Overall Rating (Weighted value of above items): 4

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes

Needs more experimental results? Yes

Needs comparative evaluation? Yes

Improve critical discussion ? (validation): Yes

Figures are adequate ? (in number and quality): Yes

Conclusions/Future Work are convincing? Yes

References are up-to-date and appropriate? Yes

Paper formatting needs adjustment? Yes

Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

Advantages of this method:

This paper is the first to evaluate the dual impact of generative AI on spam detection, ultimately identifying a fundamental asymmetry between offensive and defensive capabilities. Through three distinct experimental paths, the study concludes that unless defenders proactively adopt ensemble methods and cross-model augmentation strategies, generative AI will give attackers a significant advantage.

Weaknesses of the method:

The dataset is based on CEAS-08, which features spam characteristics from 2008. These may differ from how modern LLMs interact with content. The models used are limited to SVM and Random Forest, making it difficult to generalize findings to LLM-specific models.

Some question and suggestions:

1. Was there a data cleaning process for the real-world dataset?
2. How is the quality difference of synthetic data defined?
3. Could add large language models be incorporated?
4. Consider adding modern corpora (e.g., Enron, Nazario, MeAJOR) to validate cross-corpus generalization and provide supplementary training details.
5. Include a flowchart to facilitate quick understanding of the methodology.
6. Add deep learning classifiers .

7. Semantic features could be extracted using Sentence-BERT.

Suggested improvements:

1. Avoid relying solely on the CEAS-08 dataset, as it may not represent the characteristics of modern spam.
2. Recommend providing specifications of the experimental hardware setup.
3. Incorporate existing transformer models for comparison to assess their generalization capabilities in broader cybersecurity domains (e.g., social engineering and adversarial content generation).
4. Sentence-BERT is specifically designed for sentence similarity and classification tasks, making it suitable for spam detection, sentiment analysis, and related applications.